

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
 Department of Computer Science & Engineering
 B. Sc. in CSE Semester Final Examination, Spring-2024
 Course Code: CSE-1223 Course Title: Discrete Mathematics
 Total marks: 50 Time: 2 hours 30 minutes

[Answer *all* the questions; in some questions, there are options; you will solve any one of them; Figures in the right-hand margin indicates full marks. Separate answer script must be used for Group-A and Group-B]

Group-A

CO DL

1.
 a) Proof the following theorem: 2 CO1 U
 "Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$."
 [OR]
 Show that if n is an integer greater than 1, then n can be written as the product of primes.
 b) Express $\gcd(84, 119)$ as a linear combination of 84 and 119. Show the steps. 3 CO2 App
 [OR]
 Find an inverse of 101 modulo 4620.
 c) Encrypt the message "Welcome to IIUC" by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters: 2 CO2 App

$$f(p) = (p + 14) \bmod 26$$

 [OR]
 Use the sieve of Eratosthenes to find all primes not exceeding 50.
 d) Find the value of x using the Chinese remainder theorem: 3 CO2 App

$$\begin{aligned} x &\equiv 1 \pmod{4} \\ x &\equiv 2 \pmod{5} \\ x &\equiv 3 \pmod{7} \end{aligned}$$

2.
 a) State the 1st and 2nd principles of mathematical induction. Why is mathematical induction valid? 3 CO1 U
 b) Prove that $\sqrt{3}$ is irrational by giving a proof by contradiction. 3 CO2 App
 [OR]
 Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. (An integer a is a perfect square if there is an integer b such that $a = b^2$)
 c) Use mathematical induction to prove that $3 + 3.5 + 3.5^2 + \dots + 3.5^n = (3(5^{n+1}-1))/4$ whenever n is a nonnegative integer. 4 CO2 App

Group-B

3.
 a) In a simplified version of the C programming language, variable names consist of one to three alphanumeric characters, where uppercase and lowercase letters are distinguished. The first character of a variable name must be a letter (from the 26 English letters), and subsequent characters can be letters or digits. Additionally, there are 10 reserved keywords that cannot be used as variable names. How many different variable names are there in this version of C? 4 CO2 App

$$N = K(n-1) + 1$$

- b) In a company's IT department, there are four different programming languages used: Python, Java, C++, and JavaScript. Each project team is assigned one programming language to work with. What is the minimum number of project teams required to be sure that at least ten will be assigned the same programming language? Explain. 3 CO2

- c) A bookstore has five different genres of books: fiction, non-fiction, mystery, science fiction, and fantasy. How many different ways can a customer choose eight books, selecting from these genres? Assume that only the genre of the book matters, not the specific titles or the order in which they are chosen. 3 CO2 App

[OR]

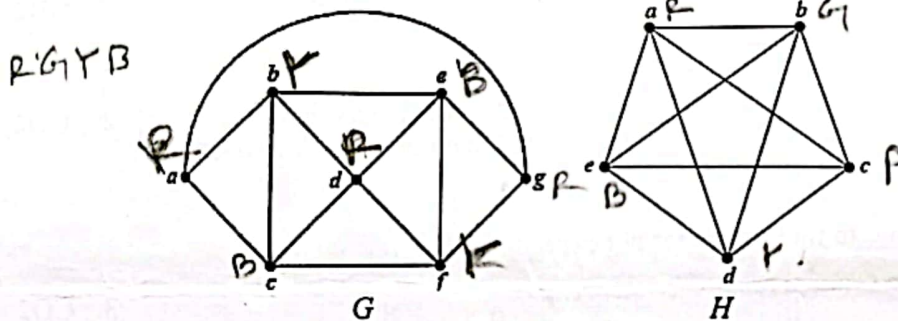
- The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain

- exactly one vowel?
- exactly two vowels?
- at least one vowel?

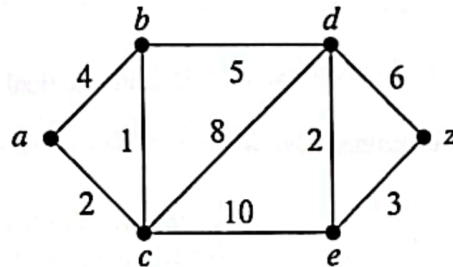
$$V_1 = \frac{1}{5} \dots$$

$$N = K(n-1) + 1$$

- a) Define Complete graph and Complete Bipartite graph. Draw the graphs: K_6 , $K_{2,6}$ 3 CO1 U
- b) State the four-color theorem. What is the chromatic number of graphs G & H? 3 CO1 U

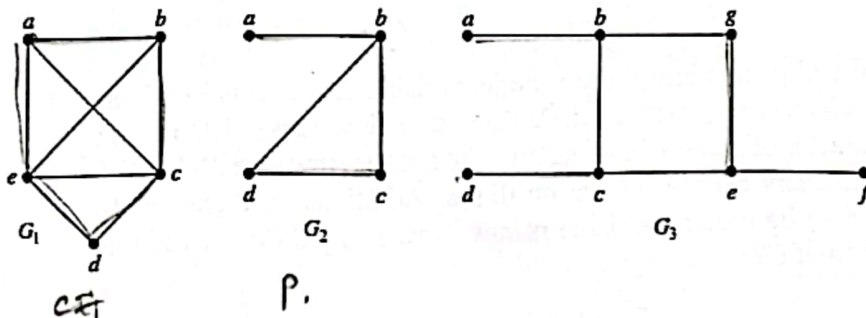


- c) Find the shortest path from a to z of the following graph using Dijkstra's algorithm. 4 CO2 App



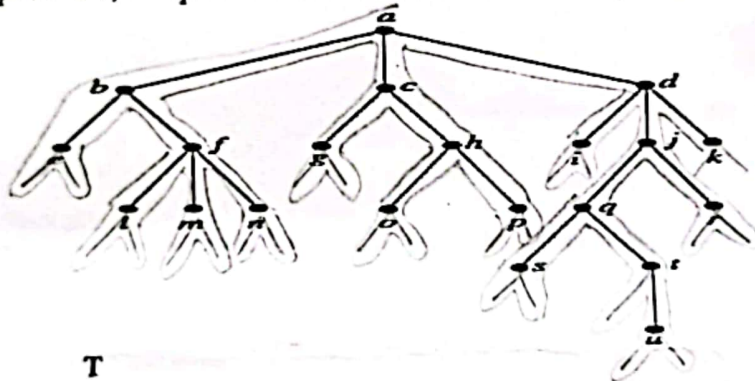
[OR]

Define Hamilton Path and Circuit. Which of the simple graphs has a Hamilton circuit or, if not, a Hamilton path?



a) Define *Tree*. Which of these G_1 and G_2 graphs are trees? Justify your answer.

Find *inorder*, *preorder*, and *postorder* traversal of the following tree T.



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[Answer *all* the questions. Figures in the right hand margin indicate full marks.
Separate answer script must be used for Group A and Group B]

GROUP-A

1. a) Suppose you are given N numbers. $a_1, a_2, a_3, \dots, a_N$ where $N \leq 10^5$ and $1 \leq a_i \leq 10^5$. You want to find their product modulo M where $M \leq 10^9$ i.e. $(a_1 * a_2 * a_3 * \dots * a_N) \bmod M$. Write a C program/algorithm to calculate the final result. 3 CO1 DL U
- b) Simulate steps of *sieve of Eratosthenes* algorithm to find all the primes not exceeding 50. 3 CO1 U

OR

- Find GCD(1000, 625) and LCM(1000, 625) using *Euclidean algorithm*.
- c) Show step by step process of finding $5^{51} \bmod 100$ using *Modular Exponentiation algorithm*. 4 CO2 Ap

OR

Find the value of x using the *Chinese remainder theorem*

$$x \equiv 2 \pmod{3}$$

$$x \equiv 1 \pmod{4}$$

$$x \equiv 3 \pmod{5}$$

2. a) Prove that $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$ whenever n is a positive integer. 3 CO2 Ap

OR

Give a *direct proof* of the theorem "If n is an odd integer, then show that $n^2 - 1$ is divisible by 8",

- b) Suppose you have an infinite number of 2 taka and 5 taka notes and you have to pay some amount of money. Use *mathematical induction* to prove that any amount greater than or equal to 4 taka can be paid with a combination of 2 taka and 5 taka notes. 4 CO2 Ap
- c) Find $f(3)$ and $f(7)$ if $f(n)$ is defined recursively by $f(0) = f(1) = 1$, $f(n+1) = f(n) - f(n-1)$, and for $n \geq 1$. 3 CO2 Ap

GROUP-B

3. a) The DNA code is made up of four letters: A, C, G, and T. How many 5-element DNA sequences
- i) end with A?
 - ii) start with T and end with G?
 - iii) contain only A and T?
 - iv) do not contain C?
- 4 CO2 Ap

- b) How many ways are there for 3 penguins and 6 puffins to stand in a line so that
 i) all puffins stand together?
 ii) all penguins stand together?

OR

On each of the 22 work days in a particular month, every employee of a start-up venture was sent a company communication. If a total of 4642 total company communications were sent, how many employees does the company have, assuming that no staffing changes were made that month?

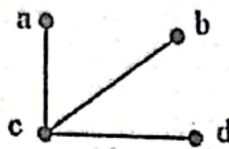
- c) State the *pigeonhole principle*. Use pigeon hole principle to show that among any group of 20 people (where any two people are either friends or enemies), there are either **four** mutual friends or **four** mutual enemies.

OR

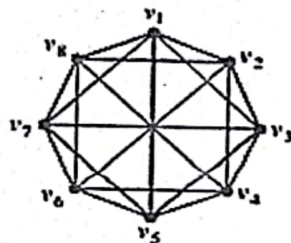
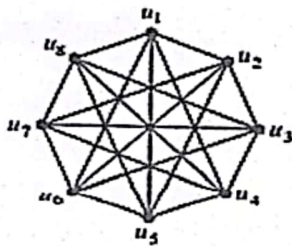
How many positive integers between 5 and 31

- i) are divisible by 3? Which integers are these?
 ii) are divisible by 4? Which integers are these?
 iii) are divisible by 3 and by 4? Which integers are these?

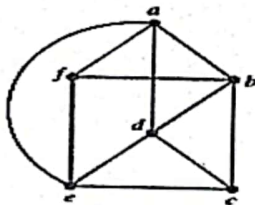
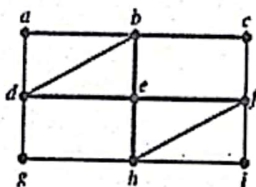
4. a) State the *Handshaking theorem*. Verify the Handshaking theorem for the following figure.



- b) Are the graphs *isomorphic* or not? Please justify your answer.

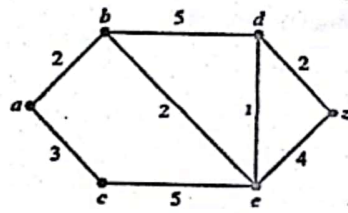


- c) Determine whether the given graph has a *Euler circuit*. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has a *Euler path* and construct such a path if one exists.



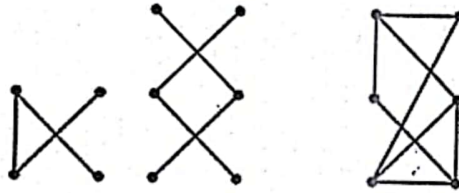
OR

Apply *Dijkstra's algorithm* for finding the shortest paths between a to z.



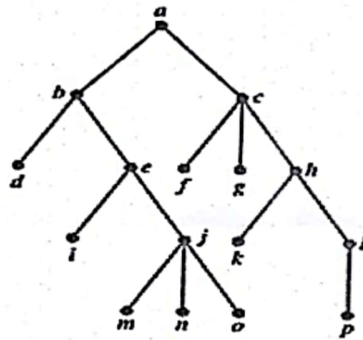
5. a) Which of this graph is *tree*? Justify your answer.

3 CO1 U



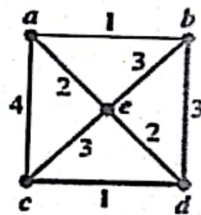
- b) Find *inorder*, *preorder*, and *postorder* traversal of the following tree T.

3 CO1 U



- c) Define *minimum spanning tree*. Use *Kruskal's* or *Prim's* algorithm to find a minimum spanning tree for the given weighted graph

4 CO2 Ap



Department of Computer Science and Engineering

Semester: 2nd

Course Title: Discrete Mathematics

Marks: 50

- (i) The figures in the right-hand margin indicate full marks**

Course Outcomes (COs) of the Questions	
CO1	Understand fundamental concept of different discrete structures like set, function, relation, graph, tree etc. and their properties. Also, the concept of different types of formal logic and mathematical reasoning, Graphs and trees
CO2	Apply the concept of formal logic, mathematical reasoning various concepts of number theory and combinatorics, Graphs and trees

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

GROUP-A

- Q1.**

a) Find all incongruent solutions to the following: i) $7x \equiv 3(\text{mod } 15)$ ii) $x^2 \equiv 1(\text{mod } 8)$	<table border="0"> <tr> <td></td> <td style="text-align: center;">CO</td> <td style="text-align: center;">DL</td> </tr> <tr> <td style="text-align: right;">4</td> <td style="text-align: center;">CO1</td> <td style="text-align: center;">C3</td> </tr> </table>		CO	DL	4	CO1	C3
	CO	DL					
4	CO1	C3					

OR

Find value of x using Chinese remainder theorem

$$x \equiv 1 \pmod{2}$$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

- b) Find the highest common factor of 34, 42 and 58.

OR

Find the prime factorization of 118724.

- c) Express $\gcd(252, 198)$ as a linear combination of 252 and 198. Show the steps. 3 CO2 C3

- | | | | |
|--|---|-----|----|
| d) Write down the applications of Congruence | 1 | CO1 | C2 |
|--|---|-----|----|

- Q2. a) Prove that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ 4 CO2 C3

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

Whenever n is nonnegative integer

OR

OR
Use mathematical induction to show that 3 divides n^3+2n whenever n is a positive integer.

- b) Let $P(n)$ be the statement that $1^3 + 2^3 + \dots + n^3 = (n(n+1)/2)^2$ for the positive integer n . 6 CO2 C3

- What is the statement $P(1)$?
- Show that $P(1)$ is true, completing the basis step of the proof of $P(n)$ for all positive integers n
- What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ?
- What do you need to prove in the inductive step of a proof that $P(n)$ is true for all positive integers n ?
- Complete the inductive step of a proof that $P(n)$ is true for all positive integers n , identifying where you use the inductive hypothesis.
- Explain why these steps show that this formula is true whenever n is a positive integer

GROUP-B

- Q3. a) State the generalized pigeonhole principle. What is the minimum of students required in a Discrete Mathematics class to be sure that at least eight will receive the same grade, if there are four possible grades, namely A+, A, A- and F. 3 CO2 C2

- b) Write some applications of Sum and-product rule. 2 CO1 C2

- c) The "hello, world" problem for the pigeonhole principle is the "sock problem": In your dresser drawer you have a jumble of socks in two colors, say blue and gray. It's dark, and you don't want to wake your spouse. 3 CO2 C3
- How many socks must you grab to guarantee that you have a pair of the same color?
 - How many socks must you grab to guarantee that you have 3 socks of the same color?

Or,

You have pens of many colors. In the drawer there are 43 black pens, 2 red pens, 23 blue pens and 8 orange pens. What is the minimum numbers of pens you need to take off from the drawer in order to be certain that you have taken four pens of same colors?

- d) How many different license plates can be made if each plate contains a sequence of three uppercase English letter followed by three digits? 2 CO1 C2

Or,

How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

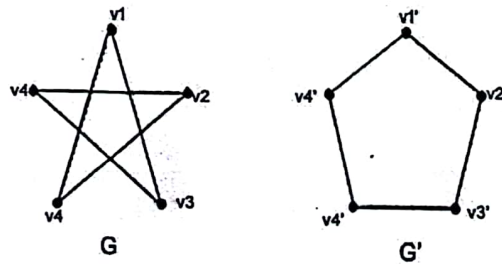
- Q4. a) Define with example: 2 CO1 C1

- i) Complete graph ii) Bipartite Graph

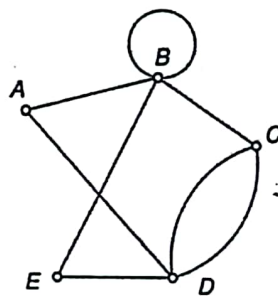
Or,

Draw Two graph K_5 , $K_{3,4}$

- b) What do you mean by isomorphism of graphs? Determine whether the following two graphs are isomorphic or not. 3 CO2 C3

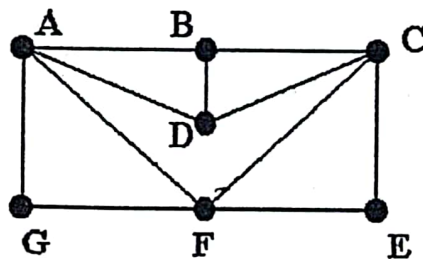


- c) Determine whether the following graph has a Euler circuits or path yes or not? Construct such a circuit or path if exists. 3 CO3 C3

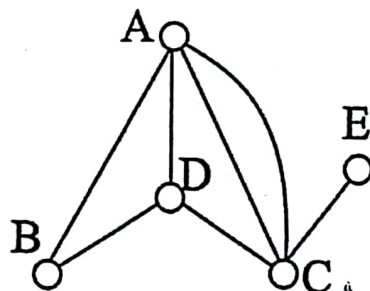


Or

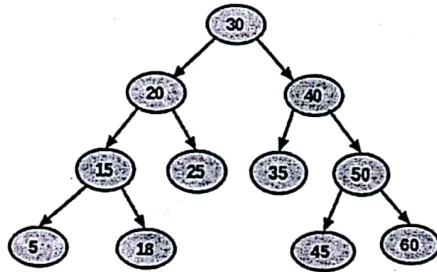
Determine whether the following graph has a Hamiltonian circuits or path yes or not? Construct such a circuit or path if exists.



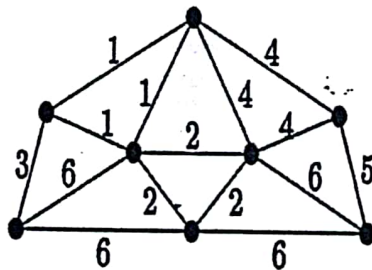
- d) Find adjacency list and matrix of the following graph. 2 CO2 C2



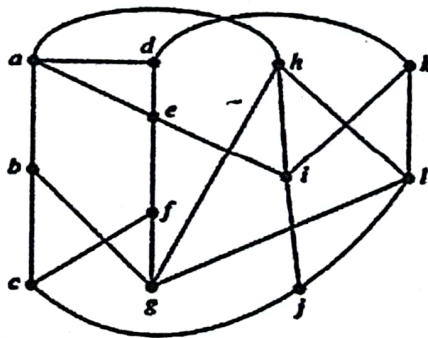
- Q5 a) What do you mean by tree traversing? Find inorder, preorder and postorder traversal of the following tree. 3 CO3 C2



- b) Define minimum spanning. Construct a minimum spanning tree of the following graph. 4 CO2 C3



- c) What is a spanning tree? How many edges must be removed from a connected graph with n vertices and m edges to produce a spanning tree? Find a spanning tree for the following graph. 3 CO2 C3





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Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
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Group -A

CO DL

Answer the following Questions

- a) Prove that if a and b are nonzero integers, a divides b , and $a + b$ is odd, then a is odd. 2 CO2 App
Or
Prove that if a is a positive integer, then 4 does not divide $a^2 + 2$
- b) Find each of these values. 2 CO2 App
a) $(177 \bmod 31 + 270 \bmod 31) \bmod 31$
b) $(177 \bmod 31 \cdot 270 \bmod 31) \bmod 31$
- c) Write down the differences between GCD and LCM? The answer should include appropriate example for both types. 3 CO1 U
Or
Define Linear Congruence. What are the solutions of the linear congruence $3x \equiv 4 \pmod{7}$
- d) Find x using Chinese remainder theorem, if possible, such that 3 CO2 App
 $2x \equiv 5 \pmod{7}$, and
 $3x \equiv 4 \pmod{8}$
- a) Write the 1st and 2nd principle of mathematical induction. Why mathematical induction is valid? 3 CO1 R
- b) Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer. 3 CO3 App
Or
Use mathematical induction to prove that the sum of the first n positive odd integers is n^2 .

- c) Explain Chinese Remainder Theorem with example.

4 CO2 U

Or

Show that if n is an integer greater than 1, then n can be written as the product of primes

Group -B

Answer the following Questions

- a) How many different three-letter initials with none of the letters repeated can people have?

2 CO2 U

- b) How many bit strings are there of length six or less, not counting the empty string?

2 CO1 U

- c) A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.

3 CO2 U

i) How many balls must she select to be sure of having at least three balls of the same color?

ii) How many balls must she select to be sure of having at least three blue balls?

Or

Suppose a license plate containing three letters followed by two digits with the last digit not zero. How many different license plates can be printed?

- d) State the pigeonhole principle. Explain how pigeonhole principle can be used to show that among any 11 integers, at least two must have the same last digit.

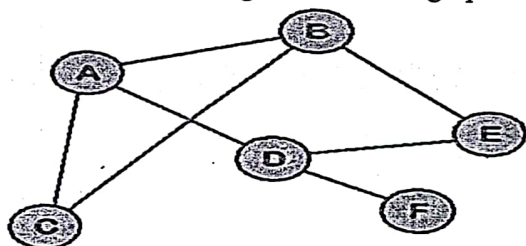
3 CO1 U

- a) Define with example: simple graph and bipartite graph.

2 CO1 R

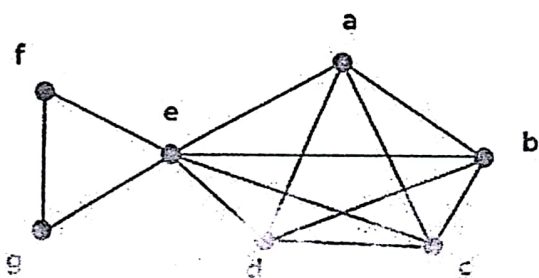
- b) State the Handshaking theorem of a graph. Verify this theorem for the following graph.

2 CO2 U



- c) Find the Euler path and circuit of the following graph

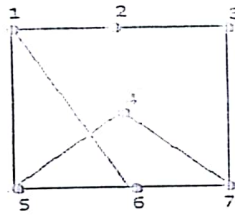
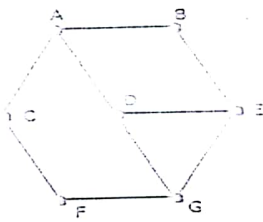
3 CO3 App



Or

What do you mean by isomorphism of graphs? Determine whether the given pair of

graphs in the following figure is isomorphic.



- d) What do you mean by Incidence Matrices? Draw the graph by the following Incidence Matrices. 3 CO2 U

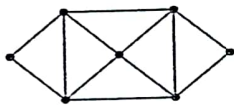
$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

- a) Explain why every tree is a bipartite graph. 2 CO1 U

Or

Write down the differences between Graph and Tree with example.

- b) Find two different spanning trees of the graph 2 CO2 U



Or

Which of the following graphs are trees?

- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and $E = \{\{a, b\}, \{a, e\}, \{b, c\}, \{c, d\}, \{d, e\}\}$
- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and $E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{d, e\}\}$
- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and $E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{a, e\}\}$
- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and $E = \{\{a, b\}, \{a, c\}, \{d, e\}\}$

- c) For each degree sequence below, decide whether it must always, must never, or could possibly be a degree sequence for a tree. Remember, a degree sequence lists out the degrees (number of edges incident to the vertex) of all the vertices in a graph in non-increasing order. 3 CO2 App

a. (4,1,1,1)

b. (3,3,2,1,1)

c. (2,2,2,1,1)

d. (4,4,3,3,3,2,2,1,1,1,1,1,1)

- d) Write down the differences between prim's and kruskal algorithms with example. 3 CO1 U

Or

Define with figure: graph coloring, simple graph and weighted graph.



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Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group -A
Answer the following questions

1. a) Define Linear Congruence. What are the solutions of the linear congruence $3x \equiv 4 \pmod{7}$ 3 CO2 DL U
- b) Find the GCD of 111 and 201 using Euclidian algorithm. 2 CO2 App
- c) What do you know pseudorandom number? Find the sequence of 8 pseudorandom number by choosing $m=17$, $a=5$, $c=2$ and $x_0=3$. 3 CO2 APP
 Or
 Find the smallest positive integer x such that
 $x \equiv 1 \pmod{5}$
 $x \equiv 2 \pmod{7}$
 $x \equiv 3 \pmod{9}$
 $x \equiv 4 \pmod{11}$
- d) Prove, for all integer $n \geq 4$, $2^n < n!$ 2 CO1 R
2. a) Write the 1st and 2nd principle of mathematical induction .Why mathematical induction is valid? 3 CO1 R
- b) Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer. 3 CO3 App
 Or
 Use mathematical induction to prove that the sum of the first n positive odd integers is n^2 .
- c) Explain Chinese Remainder Theorem with example. 4 CO2 U
 Or
 Show that if n is an integer greater than 1, then n can be written as the product of primes

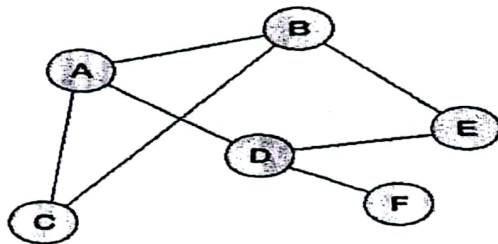
Group -B
Answer the following questions

3. a) What do you about principles of counting? Discuss sum rule and product rule with an example. 2 CO1 R
- b) In how many ways Computer Club containing 50 members elect a General secretary, assistant General secretary and an executive(assuming no person is elected to more than one position) 3 CO2 U

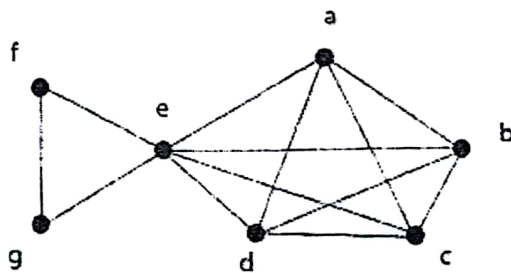
Or

Suppose a license plate containing three letters followed by two digits with the last digit not zero. How many different license plates can be printed?

- c) Describe Inclusion-Exclusion principle with an example. 2 CO2 U
- d) State the pigeonhole principle. Explain how pigeonhole principle can be used to show that among any 11 integers, at least two must have the same 3 CO2 U
4. a) Define with example: i) simple graph and ii) bipartite graph. 2 CO1 R
- b) State the Handshaking theorem of graph. Verify this theorem for the following graph. 2 CO2 U

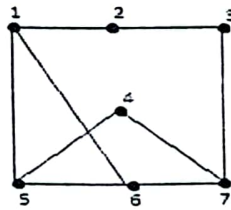
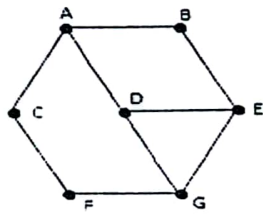


- c) Find the Euler path and circuit of the following graph 3 CO3 App



Or

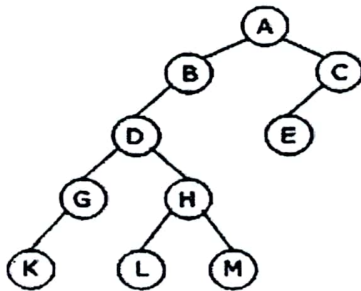
What do you mean by isomorphism of graphs? Determine whether the given pair of graphs in the following figure is isomorphic.



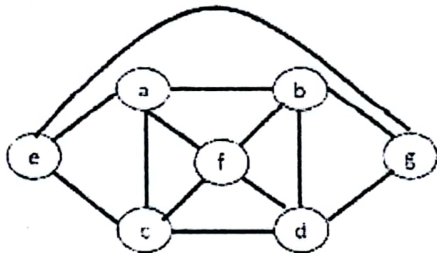
- d) What do you mean by Incidence Matrices? Draw the graph by the following Incidence Matrices. 3 CO2 U

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

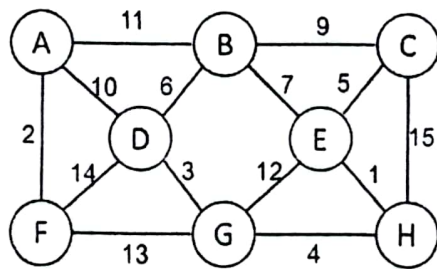
- 5 a) What do you mean by tree traversing? Find the inorder, preorder and postorder traversal of the following tree 3 CO3 App



- b) What is graph coloring? Use graph coloring algorithm on the following graph 3 CO3 App



- c) Define spanning tree. Find the minimum spanning tree of the following graph 4 CO3 App



Or

Use breadth-first search to find a spanning tree for the following graph.

